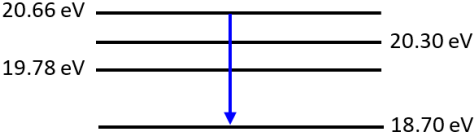


6	(a)		A photon is a quantum (packet) of electromagnetic radiation / wave / field.	
	(b)		$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{6.33 \times 10^{-7}} = 3.14 \times 10^{-19} \text{ J}$	
	(c)		$P = \frac{Nhf}{t} \quad \frac{N}{t} = \frac{P}{hf} = \frac{1 \times 10^{-3}}{3.14 \times 10^{-19}} = 3.2 \times 10^{15}$	

	(d)	 Fig 6.2	
	(e)	(i) UP	
		(ii) Number of photoelectrons emitted per unit time $= \frac{20}{100} \times 3 \times 10^{15}$ $= 6 \times 10^{14}$ Current, $I = \frac{Ne}{t} = 6 \times 10^{14} \times 1.6 \times 10^{-19}$ $= 9.6 \times 10^{-5} \text{ A}$	
		(ii) Most of the incident photons are reflected by the photodiode.	